

FleetFlow

Software Requirements Specification (SRS)

1. Introduction

- 1.1 Purpose: FleetFlow is a web-based transport operations management system that centralizes day-to-day fleet activities, including vehicles, drivers, trips, maintenance, fuel, expenses and operational monitoring.
- 1.2 Scope: The system covers the workflow from authenticated access and resource management through trip dispatch/completion and post-trip maintenance, fuel, expense and dashboard updates. Validation rules prevent invalid assignments and inconsistent fleet states.
- 1.3 Intended Audience: Fleet managers, drivers, safety/compliance personnel, financial analysts, developers and testers.
- 1.4 References: FleetFlow project document; TransitOps Smart Transport Operations Platform specification; Software Requirements Specification template.

2. Overall Description

- 2.1 Product Perspective: FleetFlow is a centralized replacement for spreadsheet/manual-logbook based transport operations, intended to reduce scheduling conflicts, underutilized vehicles, missed maintenance, licence-compliance issues, inaccurate expenses and poor operational visibility.
- 2.2 Major Features:
- Authentication and role-based access control (RBAC).
 - Vehicle and driver registry with availability/status tracking.
 - Trip creation, dispatch, completion and cancellation.
 - Maintenance workflow linked to vehicle availability.
 - Fuel and operational expense recording.
 - Dashboard KPIs and operational analytics.
- 2.3 User Classes: Fleet Manager manages fleet assets, maintenance and operations; Driver performs trip-related activities; Safety Officer monitors licence validity and safety/compliance; Financial Analyst reviews fuel, maintenance, expenses and profitability.
- 2.4 Operating Environment: Responsive web application using a React/Vite frontend, Express/Node.js backend, PostgreSQL database and HTTP/JSON API communication.
- 2.5 Constraints and Assumptions: Protected functionality requires authentication; RBAC controls access; vehicle/driver availability and licence validity must be enforced; critical state changes must remain consistent with database records.
- 2.6 Agile Development Approach:** FleetFlow will follow the Agile Software Development Life Cycle (SDLC), with short iterations that incrementally deliver, test and integrate authentication, fleet/driver management, trip management, maintenance, fuel/expense tracking and dashboard analytics. Requirements may be refined between iterations through review and feedback while preserving core business rules.

3. Functional Requirements

Requirements will be prioritized and implemented iteratively; high-priority operational workflows are delivered first, while lower-priority enhancements may be scheduled for later Agile iterations.

Feature	ID	Requirement
Authentication & Authorization	FR-01	Users shall log in using email/password; unauthenticated users shall be denied protected access; permissions shall be role-based.
Vehicle Management	FR-02	The system shall create, update and view vehicles; registration number shall be unique; status shall support Available, On Trip, In Shop and Retired.
Driver Management	FR-03	The system shall maintain driver identity, licence, expiry, contact, safety score and status; duplicate licence numbers and invalid assignments shall be rejected.
Trip Management	FR-04	Users shall create trips with source, destination, vehicle, driver, cargo weight and planned distance; lifecycle shall be Draft → Dispatched → Completed/Cancelled.
Maintenance	FR-05	The system shall create and close maintenance records; an active record shall set the vehicle to In Shop and remove it from dispatch selection.
Fuel & Expenses	FR-06	The system shall record fuel and other operational expenses and reject invalid/negative amounts; vehicle operating cost shall be calculated from recorded costs.
Dashboard & Analytics	FR-07	The dashboard shall present fleet KPIs and update operational metrics when fleet, trip, maintenance, fuel or expense data changes.

4. Mandatory Business Rules

- BR-01: Retired or In Shop vehicles shall never appear in dispatch selection.
- BR-02: Drivers with expired licences or Suspended status shall not be assigned to trips.
- BR-03: A vehicle or driver already On Trip shall not be assigned to another active trip.
- BR-04: Cargo weight shall not exceed the selected vehicle maximum load capacity.

- BR-05: Dispatching a trip shall set both vehicle and driver status to On Trip.
- BR-06: Completing or cancelling a dispatched trip shall restore vehicle and driver availability.
- BR-07: Opening maintenance shall set a vehicle to In Shop; closing maintenance shall restore Available unless the vehicle is Retired.

5. External Interfaces

5.1 User Interface: Responsive web screens for login, dashboard, vehicles, drivers, trips, maintenance, fuel and expenses, with forms, tables, status indicators, validation and error feedback.

5.2 Software Interface: React/Vite frontend communicates with an Express/Node.js REST API; PostgreSQL stores users, vehicles, drivers, trips, maintenance logs, fuel logs and expenses.

5.3 Communications: HTTP/HTTPS requests with JSON payloads; authenticated API access for protected operations.

6. Non-Functional Requirements

- Performance: Normal CRUD, dispatch and dashboard operations should respond promptly and reflect committed database changes.
- Security: Authentication, RBAC and protected routes/APIs shall prevent unauthorized access to fleet data and operations.
- Reliability & Data Integrity: Invalid assignments shall be rejected and trip/vehicle/driver status transitions shall remain consistent.
- Usability: Navigation, forms, validations, status messages and responsive layouts shall support straightforward day-to-day operations.
- Maintainability: Functional areas shall remain modular so features can be implemented, tested, integrated and refined across Agile iterations with minimal impact on other modules.

7. Database & Analysis Models

Core database entities: Users, Vehicles, Drivers, Trips, Maintenance Logs, Fuel Logs and Expenses. Recommended supporting analysis models for the full SRS are a Use Case Diagram, ER Diagram and Trip/Vehicle/Driver State Transition Diagram.